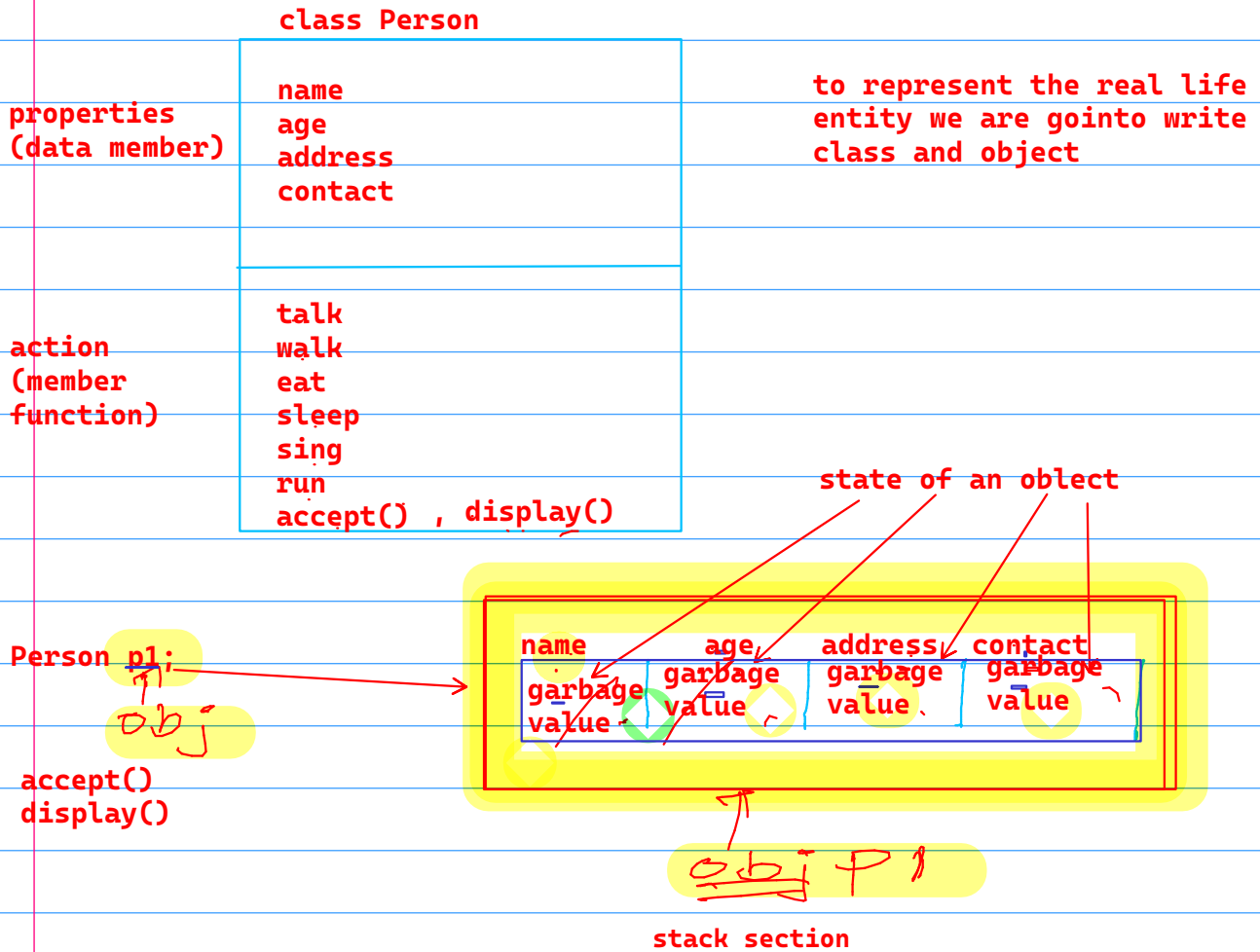
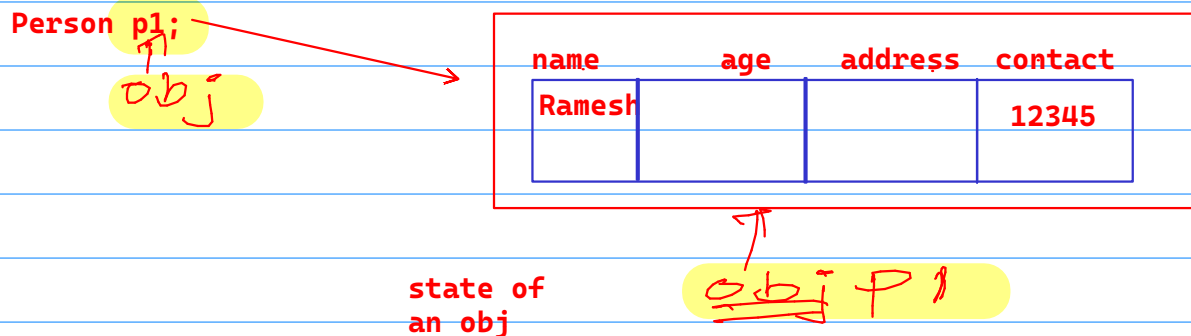


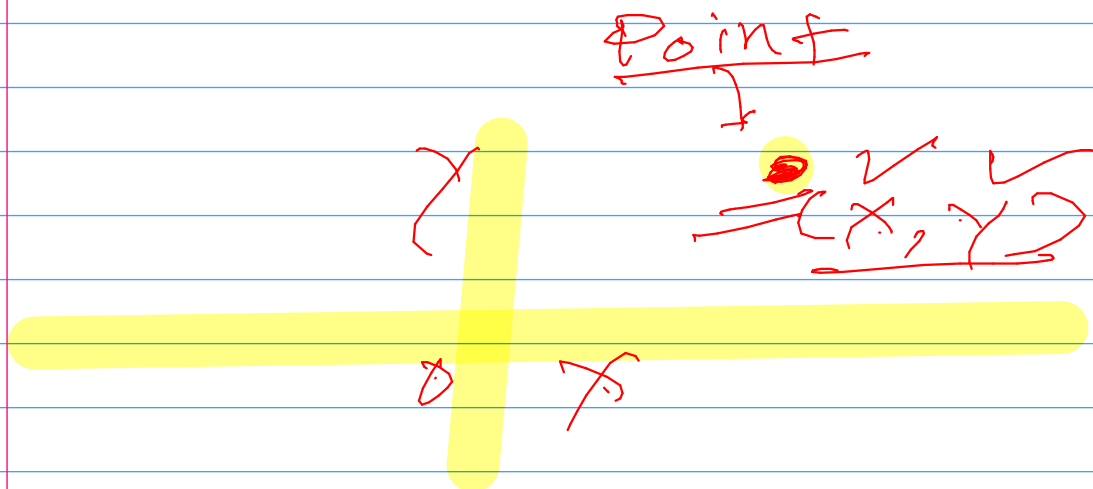
Class diagram



how to initialize an object of Person class?

Aim → Initialize an obj of class with desired values





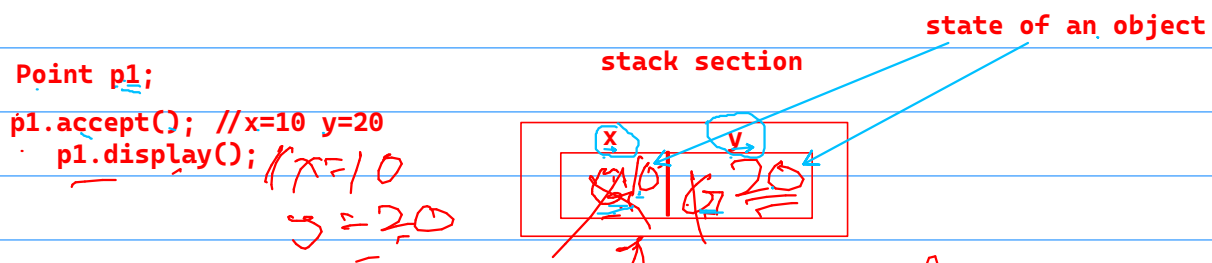
class Point

property/data member
x
y
action/ member function
accept()
display()
draw()

Point p1;

p1.accept(); //x=10 y=20

p1.display();



int num1;

num1 = 10; // assignment

int num2 = 20; // Initialization

```

class Point
{
    int x;
    int y;
    public:
        //constructor
        Point()
        {
            //initialization of data member of class
            x=10;
            y=20;
            cout<<"Inside the constructor of a class Point ... !"<<endl;
        }
}

int main()
{
    Point p1;
}

```

stack section

x	y
10	20

```

class Person
{
    private:
        string name; //property ⇒ data member of class person
        int age; //property ⇒ data member of class person
        string add; //property ⇒ data member of class person
    public:
        //Parameterless Constructor
        Person()
        {
            //this = this pointer will contain an address of p1 obj
            //In cotructor init the data membebr of class
            name="Om";
            age=23;
            add="Pune";
            cout<<this; //100
            cout<<"Inside the Parameterless constructor of Person class ... !"<<endl;
        }

        //Parameterized constructor
        Person(string nm, int a, string ad)
        {
            name=nm; //Ramesh
            age=a; //25
            add=ad; //Mumbai
            cout<<"Inside the parameterized constructor of Person class ... !"<<endl;
        }

        void display(){ }
}; //end of class

int main()
{
    Person p1; // here constructor of Person class will be called and an address if
    //p1 obj will be passed to the constructor of class automatically(implicitly)//p1 is an obj of class Perosn
    cout<<"address of p1="<<&p1<<endl; //100 // address of p1 obj
    p1.display(); //here address of obj p1 will be passed to the display() in the
    //form of this pointer automatically
}

```